

1-A:

Pareto efficiency, also known as Pareto optimality, is achieved when **it is impossible to make one person better off without making someone else worse off**. This represents that we are operating on the potential of the PPC.

In other words, it is a “mix of productive and allocative efficiency”

- Productive efficiency: Resources are being used in the most efficient way possible to produce the maximum output.
- Allocative efficiency: Resources are being allocated in a way that maximizes overall satisfaction or welfare.
- B) Equal distribution of goods and services is not necessarily Pareto efficient, as equity and efficiency are separate concepts.
- C) Social benefits being greater than private benefits is related to positive externalities, not Pareto efficiency.
- D) Social marginal benefit being greater than social marginal cost is related to the optimality of a single decision, not overall Pareto efficiency. This is because to check if the output is socially optimal, we need to equate MSC with MSB.

2-B:

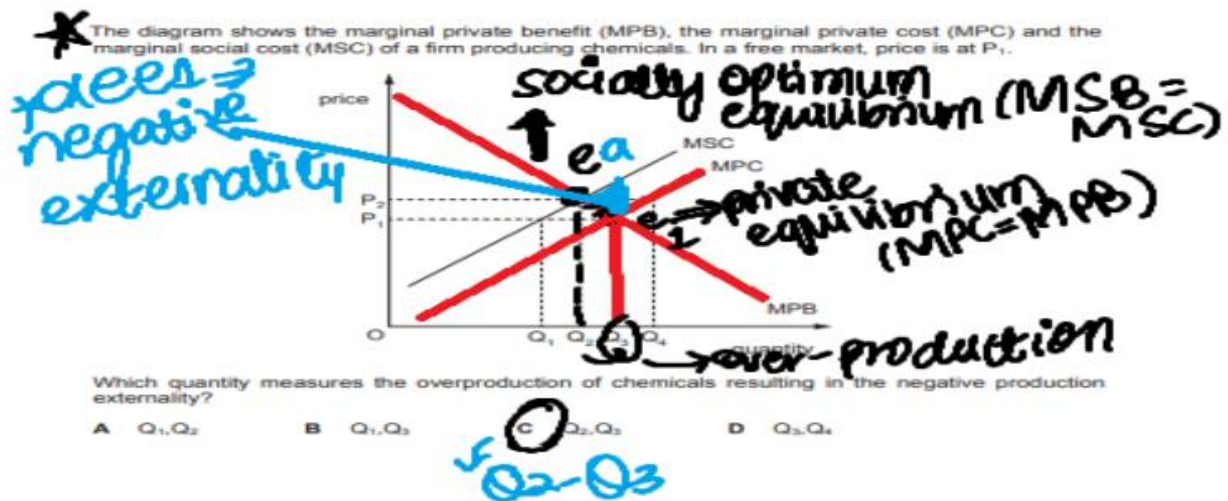
A – When the business pays compensation to other business displaced by construction work, this represents “private costs of production”. This does not represent any externality because prices or monetary values can be attached to it. Also, when foreign tourists pay to use the railway, this represents the “private benefit of consumption”. This again does not represent any externality because prices or monetary values can be attached to it.

B – When the nearby residents suffer noise from passing trains, this represents the “external cost” of this action, because the third party who is not directly involved in the transaction gets to bear it. Also, when an underground railway reduces traffic congestion, as a result of which, the air-quality improves, this will lead to fewer admissions to the hospital, which represents “external benefit”

C – Overtime payments paid by the business to its construction workers are the “private costs of production”. This again does not represent any externality because prices or monetary values can be attached to it. When overall employee productivity rises due to the construction of underground railway, which results in workers to arrive more punctually, this again represents the “external benefit”, but the overall option becomes incorrect.

D – When reduced traffic means that there are fewer road improvements needed, this represents the “external benefit”. Also, when motorists spend less on fuel, this does not represent an external benefit because prices or monetary values can be attached to this. As a result of which, this becomes incorrect.

3-C:



Q_2 represents the socially optimum level of quantity. And Q_3 represents the overproduction of the output of chemicals, because it is greater than the socially optimum level of output. This is why, key C, is correct.

4-B:

A – When the coffee is available for free, or in other words, when the price falls from 2\$ to 0\$, the qty demanded extends from X to Y, but since demand curve is the same as the marginal utility curve, it represents a situation whereby the marginal utility is zero, which means that the ‘total utility will be at its maximum’



B – As can be seen above, when the demand increases from X to Y, there comes a point where the marginal utility will be zero. This is because the demand curve=marginal utility curve. This is why, key B, is correct.

C – Jo will drink extra coffee, which is shown through an extension in qty demanded from X to Y, because it is free

D – This is incorrect because total utility falls when marginal utility is negative.

5-D:

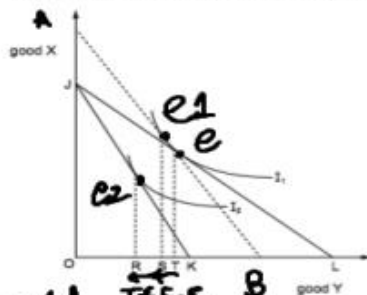
Since the inflation is rising faster than money incomes, this means that "Real-incomes" have fallen. This will result in the budget line being shifted parallelly inwards.

And, the new consumer equilibrium is likely to be point "D", where the new indifference curve is tangent to the "new budget line"

This is why, key D, becomes correct.

6-C:

The initial budget line is JL. There is then an increase in the price of good Y.



① Initial Budget line: 'JL'

② The budget line pivots inwards to 'JK'

③ In order to identify the substitution effect, we need to remove the income effect by drawing the new budget line rightwards and parallel to the old budget line (JK)

④ The old indifference curve will be tangent to the budget line 'AB' and 'JL'

⑤ The substitution effect will be from 'T-S'

The substitution effect is shown through the movement along the indifference curve, which in this case is (TS), since it results in a contraction in qty demanded from T to S.

This is why, key C, becomes correct.

7-D:

1. **Few large players:** In most countries, there are only a handful of major banks that dominate the market. This concentration of market power leads to oligopoly. Banks have some degree of market power, enabling them to influence prices, interest rates, and fees.
2. **Interdependent decision-making:** Banks closely watch their competitors' actions and adjust their own strategies accordingly. For example, if one bank lowers its interest rates, others may follow suit to remain competitive.

3. **Barriers to entry:** Starting a new bank requires significant capital, regulatory approvals, and infrastructure investments. These hurdles limit new entrants and maintain the oligopolistic structure.
4. **Product differentiation:** Banks offer various financial products and services, such as credit cards, loans, and investment products, which differ from one another. This differentiation allows banks to compete on features, prices, and services and they engage in non-price competition, such as advertising. **This is a key characteristic of a non-collusive oligopoly market structure.**

This explains why key D is correct, as it fulfills the characteristics of an “oligopoly market structure”.

While banks do link their interest rates to the central bank's rate, this doesn't mean they operate in a perfectly competitive market (A). Perfect competition assumes many firms producing identical products, which isn't the case in banking as described above. (B) Monopolistic competition (C) is also not a good fit, as some banks do have substantial market power and can influence prices. Oligopoly (D) best describes the banking industry's market structure.

8-D:

Option A: A high concentration ratio that reduces competition. This option is incorrect because a high concentration ratio, which indicates a small number of large firms dominating the industry, typically leads to less competition. It becomes challenging for small firms to enter or compete in such an environment.

Option B: High start-up costs that occur in a contestable market. This option is also incorrect. High start-up costs would typically act as a barrier to entry for small firms, making it difficult for them to establish themselves in the industry. A contestable market, on the other hand, is characterized by low barriers to entry, allowing new firms to enter and compete.

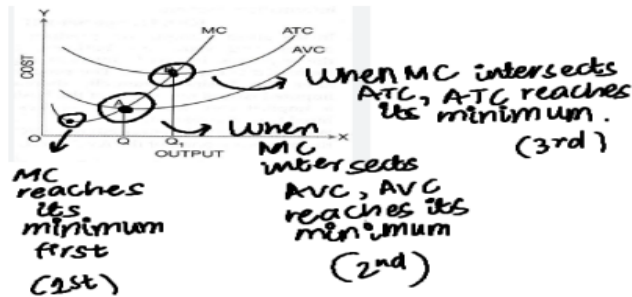
Option C: Decreasing average cost as a large firm expands. This option is incorrect because decreasing average costs as a large firm expands typically leads to economies of scale. This can create cost advantages for large firms, making it more difficult for small firms to compete due to their inability to achieve the same economies of scale.

Option D: A requirement for personalized equipment. This option is correct. The existence of smaller firms in an industry dominated by large firms can be attributed to the need for personalized equipment. Larger firms often focus on producing standardized equipment in high volumes to achieve economies of scale. However, there may be niche markets or specific customer preferences that require personalized equipment, which smaller firms can cater to. By providing specialized or customized products, small firms can carve out a niche market and differentiate themselves from the larger competitors. This is why, key D, becomes correct.

9-B:

A – This is incorrect because average fixed cost keeps on falling.

B – This is correct because the marginal cost curve intersects the average variable cost at the minimum point of the average variable cost. This is why, this becomes correct. The same goes for average total cost, as the marginal cost intersects average total cost curve at the lowest point of the average total cost curve too. This can be seen in the diagram:



C – Marginal revenue curve is a linear line.

D – And, total cost curve is an upward sloping line.

10-D:

Abnormal profits, also known as economic profits or supernormal profits, refer to the profits earned by a firm beyond what is required to keep it in business in the long run, or in other words, they represent **any amount that is greater than normal profits**. This means that the firm is earning more than the minimum necessary to cover its costs, including the opportunity costs of its resources. This explains why key D will be correct.

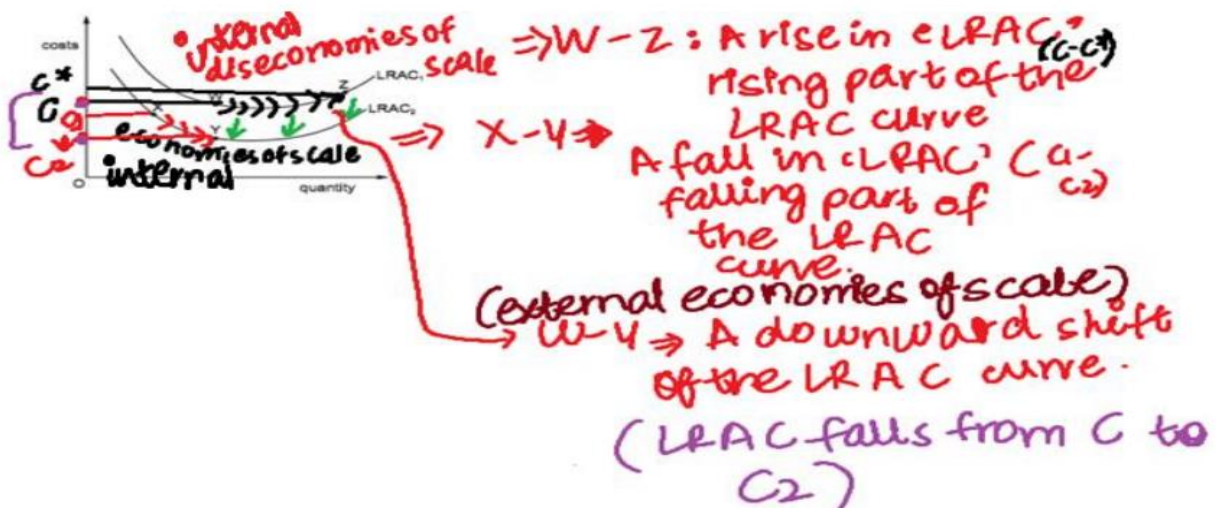
A) The extra revenue from selling land to a house-building firm is a not abnormal profit.

B) An unexpectedly large increase in demand may lead to higher revenues, but it will not necessarily lead to abnormal profits.

C) Total revenue exceeding total variable costs represents contribution.

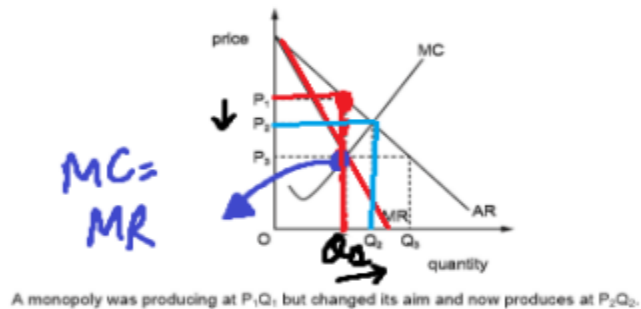
11-C:

Movement along the LRAC curve represents "internal economies of scale" & "internal diseconomies of scale". Shift of the LRAC curve indicates "external economies of scale" & "external diseconomies of scale"



12-D:

The diagram shows the cost and revenue curves for a monopoly market structure.



A – Since, the price has been lowered from P_1 to P_2 , it can be said that the firm has stock that it wants to sell, and as a result, there is an extension in the qty it sells from Q_1 to Q_2 .

B – Prices are lowered so that the new entrants are not attracted in the industry.

C – Since production has been increased, it can be said that the firms wants to benefit from “economies of scale”

D – This statement is invalid because now, the profits made will be reduced, because the firm is no longer producing at the profit-maximizing level of output (where $\text{Marginal cost} = \text{Marginal Revenue}$). Had it wanted to maximize profits, it would have stayed at the price and qty combination “ P_1Q_1 ”. As a result of which, this option becomes correct.

13-B:

A – If government intervention in form of subsidies results in “cheaper fresh fruit and vegetables”, this represents a success of the policy.

B – If government intervention in form of subsidies, when results in an “increase in production”, leads to unintended consequences such as: land clearance reducing wildlife, this represents a failure of the policy. This is why, key B, becomes correct.

C – If government intervention in form of subsidies, results in “lower spending in treating dietary illnesses”, this again represents a success of the policy.

D - If government intervention in form of subsidies, results in creation of more jobs picking fruits and vegetables, this again represents a success of the policy.

14-B:

The table shows a firm that has four possible methods of production to make its desired output. Each method has a different cost and causes a different amount of pollution.

The government rules that polluting firms must buy pollution permits that cost \$90 each. Each permit allows the firm to emit 1 tonne of pollution. No pollution can be emitted without a permit.

Which method of production has least total cost?

	cost of production (\$ per day)		pollution caused (tonnes per day)	
A	320	+	3	$3 \times 90 = 270$
B	380	+	2	$2 \times 90 = 180$
C	480	+	1	$1 \times 90 = 90$
D	700	+	0	$0 \times 90 = 0$

Handwritten calculations and notes to the right of the table:

- Red circles around the pollution cost calculations: 270, 180, 90, 0.
- Red text: "Add 590" (above 270), "560" (above 180), "570" (above 90), "700" (above 0).
- Red text: "least total cost" with an arrow pointing to the 560 value.

This is why, key B, is correct.

15-D:

Let's calculate the differences in average income between the richest 20% and the poorest 20% at each stage:

Before taxes and benefits:

Richest 20%: £85,000

Poorest 20%: £7,000

Difference: £78,000 (£85,000 - £7,000)

After all taxes (direct and indirect) and all benefits:

Richest 20%: £63,000

Poorest 20%: £17,000

Difference: £46,000 (£63,000 - £17,000)

As you can see, the difference in average income between the richest 20% and the poorest 20% decreases at each stage, indicating a reduction in income inequality due to government taxation and benefit policies. **This explains that key D will be correct.**

A) The direct taxation affected the rich more than the poor as their incomes fell by a higher amount than the amount by which average incomes of poor rose because of taxation.

B) There are still differences in incomes earned by both groups, which represents that income equality has not been achieved (avg incomes are not equal)

C) Not all income taken from the rich was redistributed to the poor, as the incomes of the poor didn't rise by the same amount.

16-B:

Transfer earnings: \$100,000 (the amount he could earn in his next best alternative job as a teacher)

Economic rent: \$400,000 (\$500,000 - \$100,000)

Transfer earnings represent the minimum amount the fashion model could earn in another job, whereas economic rent is the excess amount he earns in his current job beyond what he could earn in his next best alternative job.

In this case, the fashion model earns \$400,000 more than he would if he were a teacher, which is considered economic rent. This means that he is earning \$400,000 above his opportunity cost, which is the value of the next best alternative job he could have taken.

17-B:

A - If there is an imposition of a national maximum wage (representing a wage below the equilibrium), then in that case, the quantity demanded of labor extends, representing that there will be an increase in the number of job vacancies available.

B – Since wages are lowered because of the national maximum wage legislation, in order to make up for it and continue to attract workers, there will be an increase in other methods of rewarding television celebrities (mainly non-pecuniary or non-financial advantages)

C - There will already be a shortage of workers as because of the maximum wage legislation, quantity demanded for labor extends relative to quantity supplied for labor which contracts. This means that there cannot be a greater movement of celebrities between television channels. **Also, importantly it will be imposed across the entire labor market.**

D - Considering that QSL contracts that determines the equilibrium qty of labor or employment , there will not be an increased employment of foreign television celebrities.

18-A:

A) Subsidies for firms using recycled materials reduce costs of production, and increase Aggregate supply, thereby resulting in economic growth being experienced while promoting environmentally friendly production methods. This is why key A becomes correct.

B) Grants for oil exploration and exploitation perpetuate reliance on non-renewable resources, contributing to environmental degradation and depletion of natural resources. This option doesn't promote sustainable practices.

C) While environmental education is crucial for raising awareness and promoting environmental literacy, it **will bring a “long term impact”**. **The question asks for a faster impact.**

D) Tougher pollution controls regulate and reduce pollution, which is essential for environmental protection. However, these controls may increase costs of production for businesses, which results in Aggregate supply being reduced, as a result of which, economic growth may be hampered.

19-A:

Gross Domestic Product (GDP) is calculated as: $C + I + G + (X - M)$

→ where I represents gross investment, which does not account for depreciation.

Net National Income (NNI), on the other hand, aims to measure the net income generated by the economy. **To calculate NNI, we use net investment, which is gross investment minus depreciation.**

By subtracting depreciation from gross investment, we account for the wear and tear on capital assets, also known as fixed capital consumption. This allows us to exclude the value of capital consumed in the production process and arrive at the net income generated by the economy.

This explains why key A is correct.

B: Both GDP and NNI can be calculated using different methods (expenditure, income, or output approaches). The incomes method is not the only way to calculate NNI.

C. Double counting occurs when the value of goods and services is counted multiple times in the production process. This mainly occurs when we are calculating gdp using the output method.

D. The value of exports is already included in GDP, so it doesn't need to be added when converting to NNI.

20-D:

D) Purchasing Power Parity (PPP) is a theory that adjusts for differences in the cost of living between countries, making it a suitable method for comparing living standards across countries. **Since this is not a valid reason to question the accuracy of the data, option D becomes correct.**

The other options are valid reasons to question the accuracy of comparing living standards using Real GNP per capita:

A) Price indices may not accurately capture differences in spending patterns between countries.

B) Leisure time is an important aspect of living standards, and differences in leisure time can affect the comparison.

C) Unrecorded economies (e.g., black markets) can significantly impact the accuracy of GNP calculations, and differences in unrecorded economies between countries can lead to inaccurate comparisons.

21-D:

According to the International Labour Office (ILO) definition, unemployment refers to individuals who are:

1. **Without work (i.e., not in paid employment)**
2. **Available to start work within a short period (usually 2 weeks)**
3. **Actively seeking work (i.e., taking steps to find employment)**

The student in option D meets all three criteria, as they are:

- Without work (not in paid employment)
- Available to start work on weekends. (seeking a weekend job)
- Actively seeking work (desperately seeking a job)

The other options do not meet the criteria:

A) The single parent is not actively seeking work due to childcare constraints.

B) The retired teacher has permanently left the labor market.

C) The redundant steel worker is not actively seeking alternative work.

22-B:

B) Automation typically leads to increased labor productivity as machines and technology can perform tasks more efficiently than humans. This can free up workers to focus on higher-value tasks, leading to increased productivity as **now less labor can produce more output using less resources in less time with the help of machines. This subsequently helps to reduce labor costs too. This is why, key B, becomes correct.**

A) Encouraging women to enter the workforce can lead to an increase in labor force participation, but it may not directly lead to increased productivity per worker.

C) Raising the retirement age may lead to a decline in productivity as old age workers might not be that efficient.

D) Reducing the division of labor could lead to workers taking on a wider range of tasks, potentially decreasing productivity due to the loss of specialization efficiencies.

23-C:

$$MV = PT$$

Given that V and T are constant, any changes in the money supply (M) will directly affect the price level (P).

This means that:

- If the money supply (M) increases, the price level (P) will also increase.
- If the money supply (M) decreases, the price level (P) will decrease.

This also explains that money supply and price level is assumed to be variable.

24-B:

Note: we are interested in a two sector economy (**closed economy without a government**)

A) Aggregate demand equation is correct ($C + I$) & Multiplier's formula is incorrect (should be $1 / (1 - MPC)$, not $1/MPC$)

B) Aggregate demand equation is correct ($C + I$) & Multiplier's formula is correct ($1/MPS = 1/(1-MPC)$, since $MPS + MPC = 1$).

C) Aggregate demand equation is incorrect (should be $C + I$) & Multiplier's formula is incorrect (should be $1/(1-MPC)$, not $1/MPC$).

D) Aggregate demand equation is incorrect (should be $C + I$, not $C + I - S$) & Multiplier's formula is correct ($1/MPS$), but it's paired with the wrong aggregate demand equation.

25-B:

- Keynesians believe that aggregate demand (AD) plays a crucial role in determining economic activity, and that increasing AD can help bring the economy out of recession.
- Monetarists, on the other hand, believe that the economy has a natural tendency **to return to full employment in the long run** (LRAS is vertical), regardless of short-term fluctuations in AD.

The other options are incorrect because:

- A) Keynesians do not believe that decreasing AD leads to increased economic growth. On the contrary, monetarists believe that monetary policy is best for stimulating an economy, instead of fiscal policy.

- C) While Keynesians do believe that increasing AD can lead to increased employment, Monetarists do not believe that economic growth is best achieved solely by cutting taxation which is an expansionary fiscal policy or **Keynesian demand management policy**. **They believe in the supremacy of monetary policy.**
- D) Increasing AS (aggregate supply) is not typically seen as a solution to recession according to Keynesians. Additionally, monetarists do believe that in the long-run, the economy will be operating at its potential or in other words, there will be full employment.

26-D:

Option A states that demand-pull inflation is likely to increase. However, this is not correct. When unemployment increases, incomes decrease, leading to reduced consumption. As a result, aggregate demand decreases, which in turn reduces demand-pull inflation. Therefore, option A is incorrect.

Option B suggests that potential output is likely to increase. However, this is not true as for potential output to increase, productive capacity of the economy should increase.

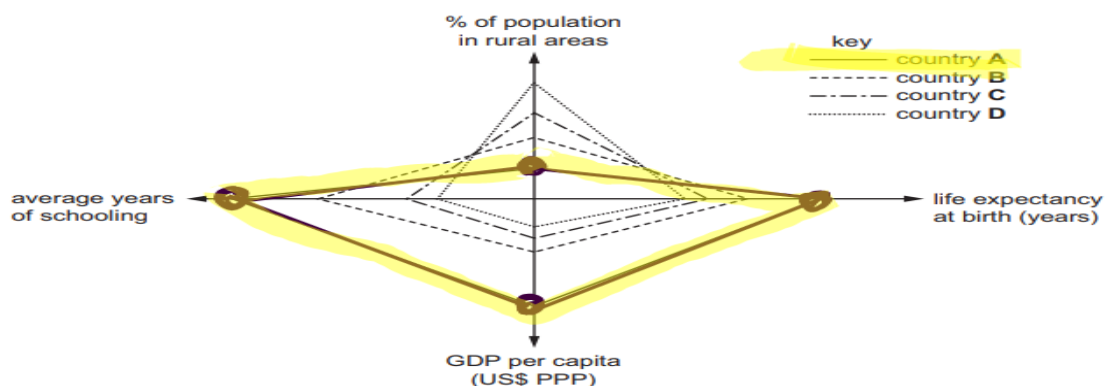
Option C states that the balance of payments deficit is likely to increase. However, the opposite is true. When incomes decrease, imports also decrease, which reduces the trade deficit (or increases the surplus). As a result, the balance of payments deficit decreases.

Option D states that the budget deficit is likely to increase. This is correct. When unemployment increases, the government receives decreased tax revenue and spends more on social welfare programs, leading to an increased budget deficit. Therefore, option D is correct.

27-A:

27 The diagram shows data on various aspects of four countries.

Which country is likely to be most developed?



The most developed country should have the least percentage of population in rural areas.

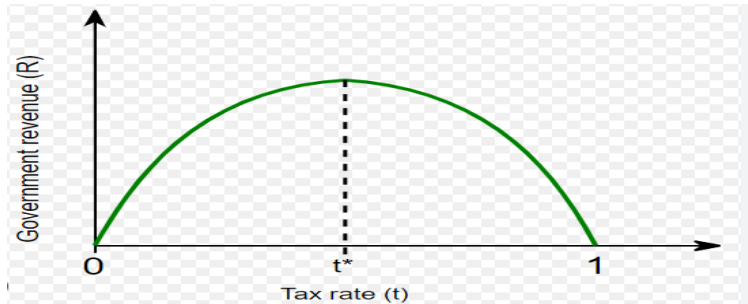
Moreover, it should also have the highest number of average years of schooling, and also, life expectancy at birth and GDP per capita should be the highest.

This is true for country A, which explains why key B will be correct.

28-D:

D) As the tax rate increases, tax revenue will eventually fall.

The Laffer curve illustrates the relationship between tax rates and government revenue. It shows that as tax rates increase, revenue initially increases, but eventually decreases as high tax rates discourage economic activity and lead to lower taxable income. **This means that beyond a certain point, further increases in tax rates will actually lead to a decrease in tax revenue. This is because of “tax evasion” i.e. when individuals and firms find loopholes to evade tax.**



The Laffer curve does not illustrate:

- A) The relationship between income and hours worked.
- B) A directly proportional relationship between income and tax revenue
- C) A constant increase in tax revenue with increasing tax rates (which is not true beyond a certain point)

29- C:

Unemployment worsens:

- Increasing direct taxation (e.g., income tax) reduces disposable income for individuals. Lower disposable incomes lead to reduced consumption, causing a decrease in aggregate demand.
- As a result, businesses may reduce production, leading to layoffs and increased unemployment as demand for labor which represents derived demand falls.

Growth worsens:

- Reduced investment and consumption (as mentioned above) lead to a fall in Aggregate demand, which represents that **actual economic growth falls**.

Balance of Payments improves:

- Higher direct taxation reduces domestic consumption and investment, leading to reduced imports (as not only people but also businesses buy fewer foreign goods and services, representing that imports of capital goods fall too).
- With reduced imports, the trade deficit (or balance of payments deficit) decreases, improving the balance of payments.

This is why, key C, becomes correct.

30-A:

- Higher income tax rates reduce disposable income for individuals.
- Reduced disposable income leads to decreased consumption and demand for goods and services. This fall in Aggregate demand relative to Aggregate Supply translates into a fall in **demand pull inflation, as the over-heated economy now "begins to cool down"**
- **This fall in Aggregate demand brings about a fall in production and subsequently demand for labor, which results in an increase in "demand-deficient unemployment"**

This explains why key A is correct.

